



1a) Solve $y = x^2$ by hand using Newton's Method, within 10^{-4} , for x val closest to (4,0)

$$P_1 = (x, x^2) \quad P_2 = (0, 1)$$

Distance between P_1 and $P_2 \Rightarrow D = \sqrt{(x-0)^2 + (x^2-1)^2}$

$$D^2 = (x-0)^2 + (x^2-1)^2 = x^4 + x^2 - 2x + 1$$

Derive Newton's Method

$$f(x) = x^4 + x^2 - 2x + 1 \quad f'(x) = 4x^3 + 2x - 2$$

See MatLab for Code.
$$X_{n+1} = X_n - \frac{f(X_n)}{f'(X_n)}$$